

Vision Transformers for Image Classification

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Evolution of Vision Feature Learning

The paper[1] talks about the evolution of vision models, starting from the CNNs to Vision Transformers.

Generally categorized in six different branches:

- Classical CNN: CiFarNet, AlexNet, GoogLeNet, Inception, Xception, DenseNet, MobileNets
- Two Stage CNN: R-CNN, Fast R-CNN, Faster R-CNN, Mask R-CNN, FPN
- One-Stage CNN: YOLO, RetinaNet, CenterNet, LADNet, SSD, and DSSD
- Autoencoders: DAE, VAE, SAE, SAE, AAE, and SSA
- Generators: GAN, InfoGAN, AC-GAN, EOSA-GAN, BiGAN, SGAN, BigGAN,
- Transformers: Swin Transformers, ViT, DETR

Vision Transformers for Image Classification[2]

Problems with original ViT:

- Large training data
- Lack of inductive biases
- High computational cost
- Balance between computational efficiency, parameter reduction and model accuracy.

Vision Transformers for Image Classification[2]

Limitations of vanilla ViT give origin to some more improved model.

They can be categorized into different key areas:

- Baseline: ViT-B, ViT-L
- Improved on Data Efficiency: DeiT, T2T ViT
- Combination with CNN: CPVT, Token-Based Visual Transformers(VT),CvT
- Deep Incorporation:CeiT
- Lite Transformers: LeViT
- Deeper Transformers: CaiT, DeepViT
- Aggregating Nested Transformers
- Cross Attention
- Patch-Wise Loss: To counter over smoothing

Vision Transformers for Image Classification[2]

Table 1: Performance comparison of various models (pre-trained) on CIFAR datasets.

Model	Parameters (M)	CIFAR-100 (%)	CIFAR-10 (%)
ViT-L/16	307.000	93.440	99.160
DeiT-B	86.000	90.800	99.100
T2T-ViT-14	21.500	87.100	97.100
CvT-W24	277.000	94.090	99.390
CeiT-S	24.200	90.800	99.100
CaiT-M36	271.000	93.300	99.300

Future Directions:

- Normalization Mechanisms: Power normalization and improved layer normalization
- Loss Mechanisms: Patch wise loss function
- Image-text training paradigms like CLIP, FPT

Vision Transformer for Image Classification Tasks[3]

Model	Base Parameter	Datasets	Accuracy(%)
ViT-B/16	86M	CIFAR-100	87.20
DeiT-B	86M	CIFAR-100	78.20
MViT-B-16	92M	CIFAR-100	76.80
Swin-B	94M	CIFAR-100	79.20
HERA	96M		
LeViT			
iGPT-L			

ViT for Remote Sensing Image Classification[4]

Model	Number of Layers	Hidden Size D	MLP Size	Heads	Number of Parameters
ViT-B	12	768	3072	12	86M
ViT-L	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Dataset	Classes	Images/Class	Resolution	Total Images
UC Merced	21	100	256x256	2100
AID	30	220-420	600x600	10,000
Optimal-31	31	60	256x256	1860

Transformer

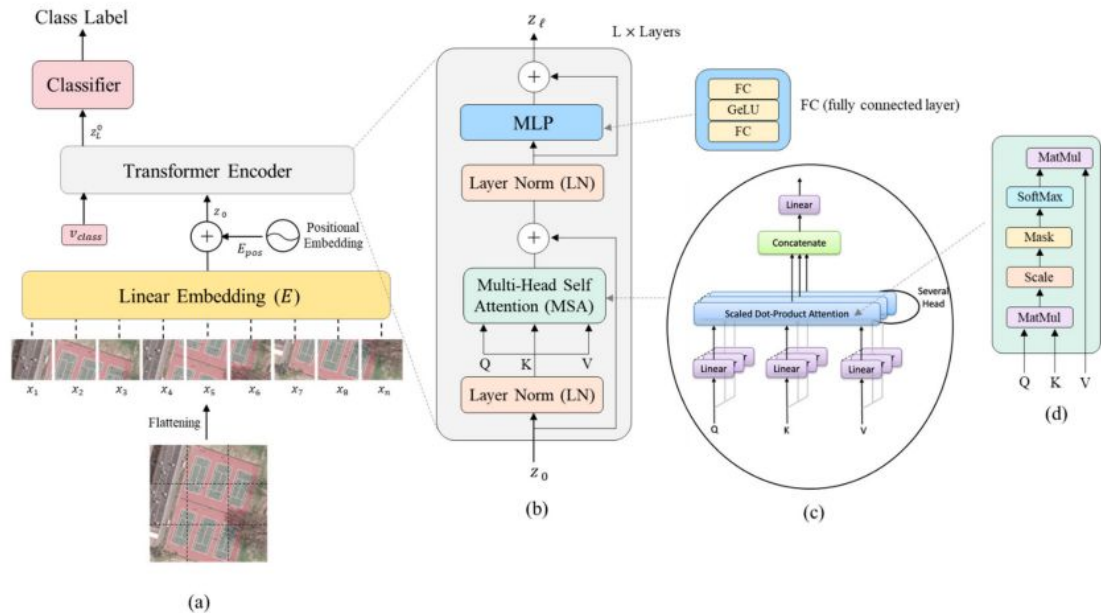


Figure 2. The Vision Transformer architecture: (a) the main architecture of the model; (b) the Transformer encoder module; (c) the Multiscale-self attention (MSA) head, and (d) the self-attention (SA) head.

Algorithm

Algorithm:

Input: Training images: $\{X_i, y_i\}_{i=1}^n$

Output: predicted labels of the test set.

1. Set batchsize to 100, Optimizer Adam (learning rate: 0.0003), number of iterations to 30, image dimensions to 224 or 384.
2. Set the number of mini-batches as: $nb = n/batchsize$
3. For iteration = 1: Number of iterations
 - 3.1 For batch = 1 : nb
 - Pick a batch from the training set,
 - Generate another batch of augmented images using a particular augmentation method,
 - Train the model on the original and augmented images by minimizing the cross-entropy loss.
 - Backpropagate the loss.
 - Update the model parameters.
4. Classify test image

Training Strategies:

1. Data Augmentation

To improve generalization, following techniques are employed:

- Standard Augmentation
- CutMix and CutOut
- Hybrid Augmentation

2. Network Compression

To analyze layer-wise learning, classification is carried out after each layer.

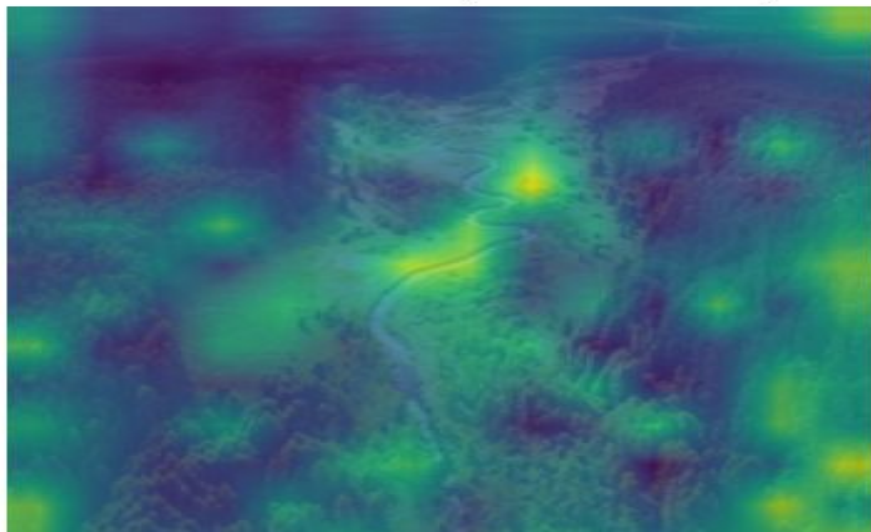
- Layer 10 seems to provide higher accuracy than Layer 12

Dataset	Clean Images	Standard	CutMix	Cutout	Hybrid	Custom
Merced(30:70)	94.55	96.32	96.66	95.44	96.73	93.88
Merced(50:50)	98.14					96.29

Top-1: river (95.39%)



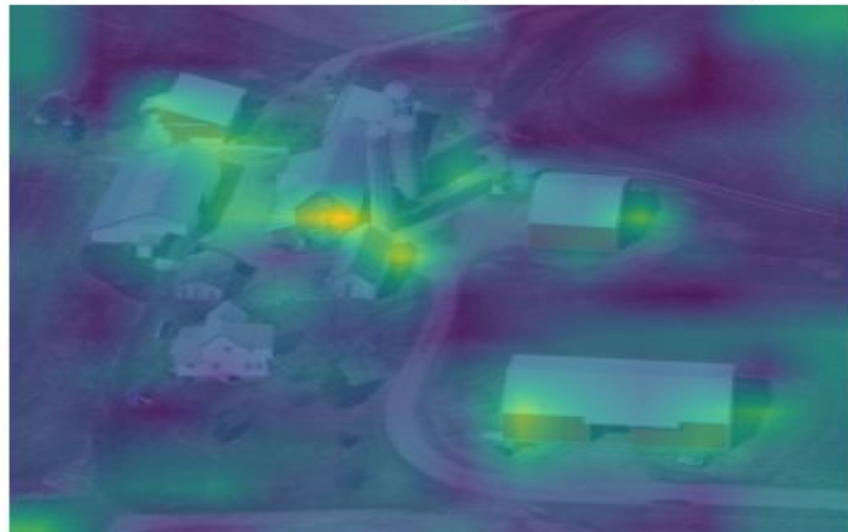
Attention Rollout (fusion=mean)



Top-1: sparseresidential (90.24%)



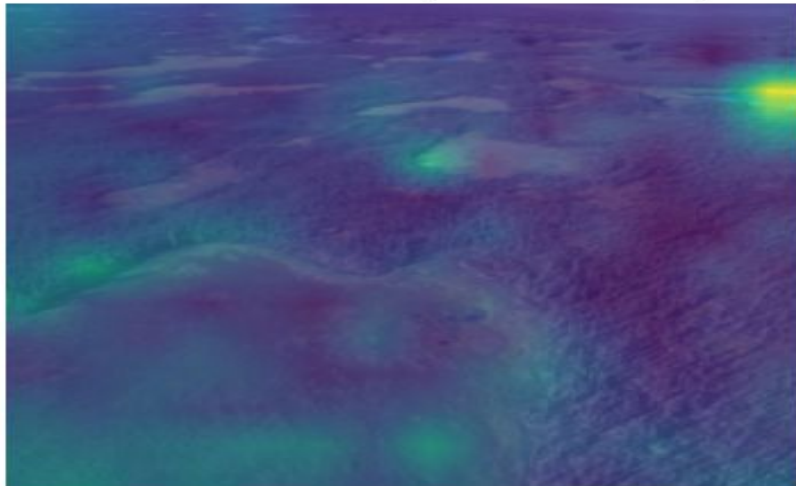
Attention Rollout (fusion=mean)



Top-1: river (99.24%)



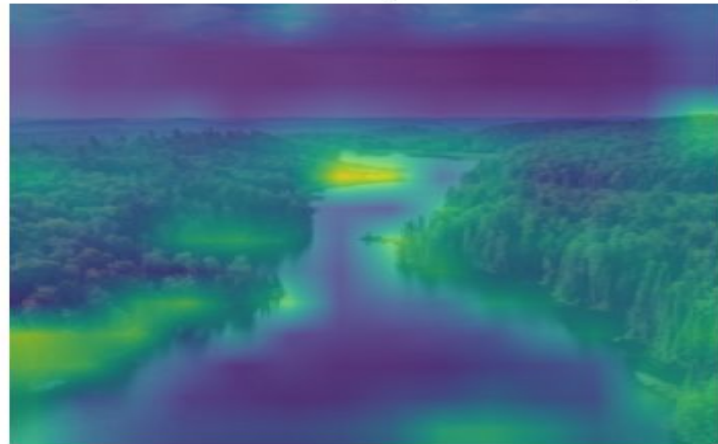
Attention Rollout (fusion=mean)



Top-1: river (93.26%)



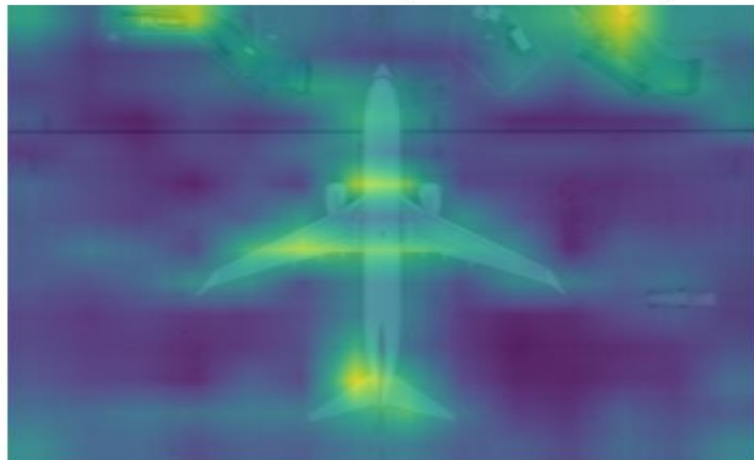
Attention Rollout (fusion=mean)



top-1: airplane (99.88%)



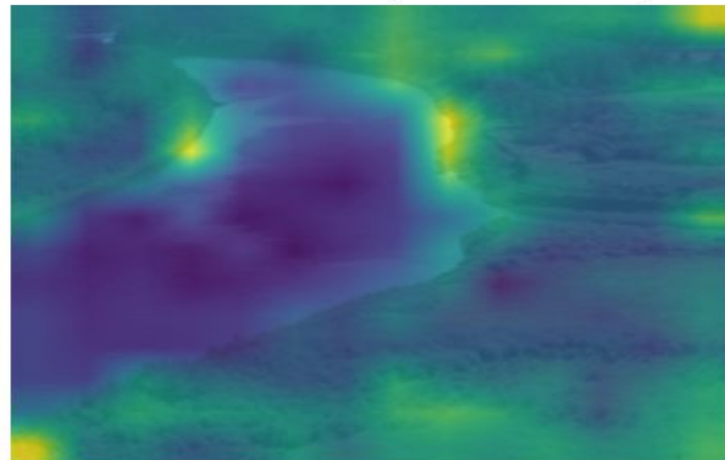
Attention Rollout (fusion=mean)



Top-1: river (99.68%)



Attention Rollout (fusion=mean)



Land Cover Image Classification[5]

Dataset: EuroSAT(27000, 64x64)

Training: 10 epochs, cross-entropy loss, Adam optimizer, lr = 1e-4

Model	Accuracy	Accuracy(Pre-Trained)
ViT32	0.917	0.972
SwinB	0.926	0.987
MaxViT	0.973	0.990

MaxViT: Multi-Axis Vision Transformer[6]

- Vanilla Transformer has quadratic complexity
- MaxViT proposes an efficient and scalable attention design called multi-axis attention
- It combines
 - Blocked local attention(captures fine local interactions)
 - Grid Attention(enable long range interaction)
- It removes Window-based attention(Swin) which can lose nonlocality as well as global attention at early resolution stage in hierarchical networks is computationally expensive

MaxViT- Architecture

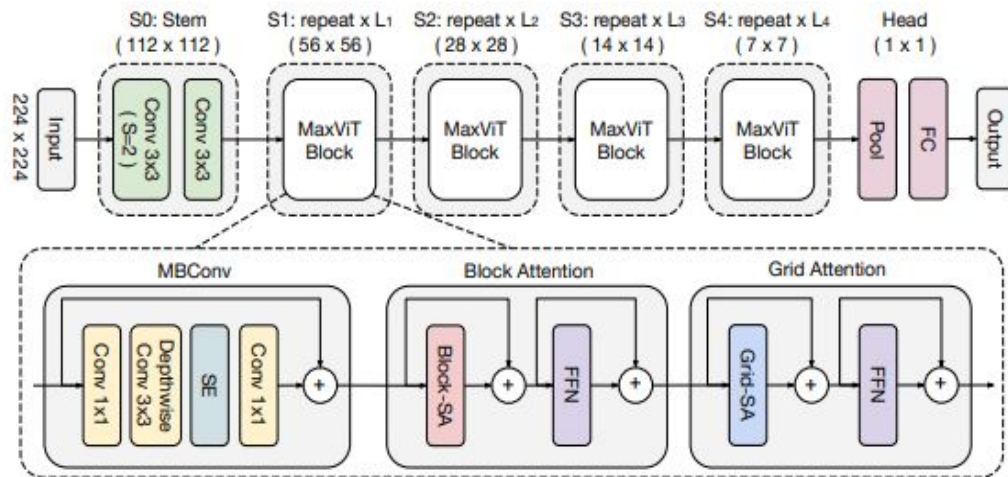


Fig. 2: **MaxViT architecture.** We follow a typical hierarchical design of ConvNet practices (e.g., ResNet) but instead build a new type of basic building block that unifies MBConv, block, and grid attention layers. Normalization and activation layers are omitted for simplicity.

MaxViT blends inductive bias (locality, translation equivariance) with multi-axis attention

- MBConv for local inductive bias
- Block(local) and Grid(global) Attention inside the transformer

Multi-axis Attention

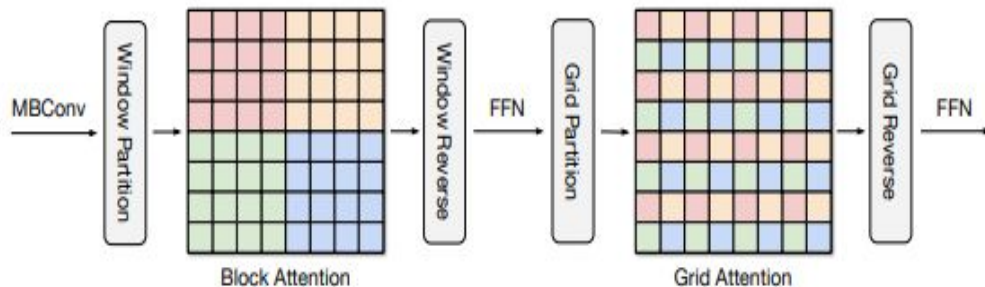


Fig. 3: **Multi-axis self-attention (Max-SA)** (best viewed in color). An illustration of the multi-axis approach for computing self-attention (window/grid size is 4×4). The block-attention module performs self-attention within windows, while the grid-attention module attends globally to pixels in a sparse, uniform grid overlaid on the entire 2D space, with both having linear complexity against input size, as we use fixed attention footage. The same colors are spatially mixed by the self-attention operation.

Block Attention(local)

- Partition the spatial map in to non-overlapping blocks
- $X \rightarrow (H/P, W/P, P, P, C)$
- Apply self attention within each block
- $O(HW.P^2)$ Linear

Grid Attention(Global, sparse)

- Partition into fixed uniform grid $G \times G$ groups, producing windows with adaptive spatial span
- $X \rightarrow (G, G, H/G, W/G, C)$
- Apply attention within each grid groups
- $O((H/G)^2 \cdot (G^2)^2)$

MBConv

MaxViT leverages the MBConv block as the main convolution operator.

Let X be the feature map from S_0 .

Feature X goes through following steps:

$X \rightarrow \text{Conv}1 \times 1 \rightarrow \text{DepthWise}(3 \times 3) \rightarrow \text{SE} \rightarrow \text{Conv}1 \times 1$

1. Without downsampling

$$x \leftarrow x + \text{Proj}(\text{SE}(\text{DWConv}(\text{Conv}(\text{Norm}(x))))))$$

2. With downsampling(stride/pooling)

$$x \leftarrow \text{Proj}(\text{Pool2D}(x)) + \text{Proj}(\text{SE}(\text{DWConv} \downarrow (\text{Conv}(\text{Norm}(x))))))$$

3. Squeeze-and-Excitation(SE)

SE reweights channels by estimating which channels are most informative.

MaxViT

Table 5: Selected ImageNet-1K comparisons (Top-1 accuracy) summarized from Tu et al. (2022).

Model	Input	Params (M)	Top-1 (%)
ViT-B/16	384×384	307.000	76.500
DeiT-B	384×384	86.000	83.100
CaiT-M24	384×384	186.000	83.400
DeepViT-L	224×224	55.000	83.100
T2T-ViT-24	224×224	64.000	82.600
Swin-B	384×384	88.000	84.500
MaxViT-L	224×224	211.000	85.170

MaxViT is also extended to downstream tasks such as **object detection** and **instance segmentation**, where it reports stronger performance than comparable hierarchical ViT backbones.

Spatial-Channel Feature Preserving Vision Transformer for Remote Sensing.(SCViT) [7]

SCViT solves 2 problems when using vanilla Transformers for Remote Sensing:

1. Under-emphasize of fine local geometric details (important for high-resolution imaginary)
2. Under-emphasize the relative contribution of channels within the classification tokens.

SCViT address this using:

1. Progressive Aggregation(PA) Module to preserve local spatial cues in token
2. Lightweight Channel Attention(LCA) module to adaptively reweight channels for classification

SCViT Architecture

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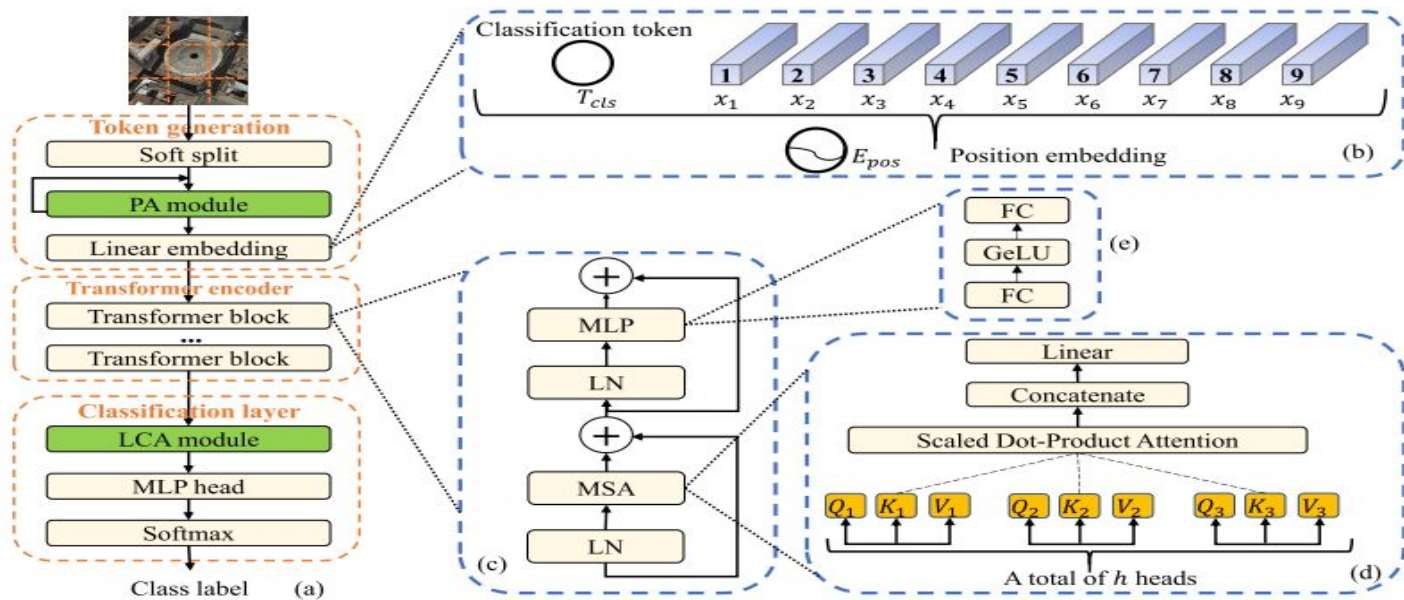
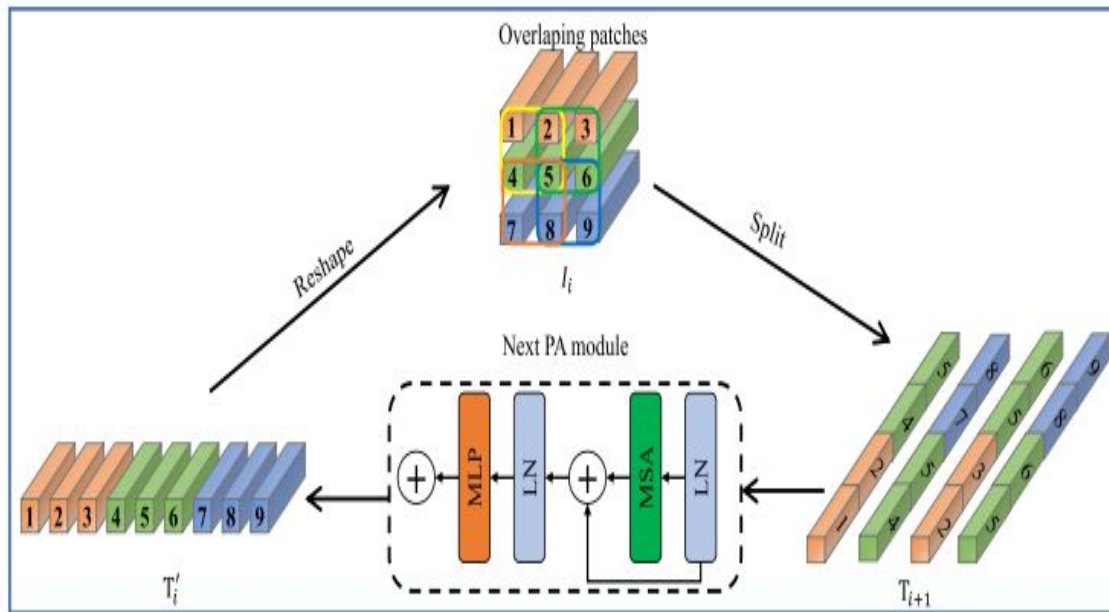


Fig. 1. (a) Proposed SCViT model consists of three main parts: token generation, the ViT encoder, and the classification layer. (b) Details of the linear embedding. (c) Overview of the ViT block. (d) MSA mechanism. (e) Structure of MLP.

Progressive Aggregation Module



The module aims to embed stronger local context into token by

- Reshape: convert token into 3D image to recover local structure
- Split with overlap: use sliding window to form overlapping patches
- Token re-aggregation: merge neighbouring patches to produce new token that preserve local context.

These token pass through standard Transformer blocks(MHSA+ MLP)

Fig. 2. PA module, including the reshape and split stages.

Classification with LCA

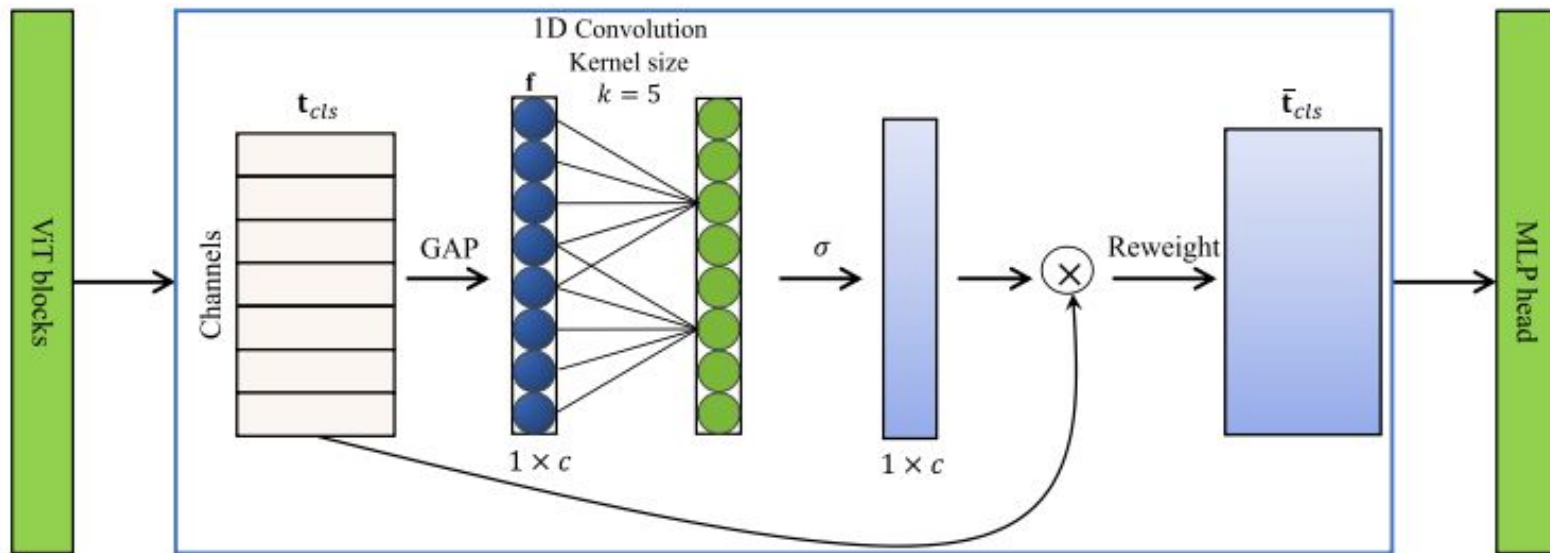


Fig. 3. Structure of the LCA module.

LCA Module

1. LCA computes the channel weights by:
 - a. Global Average Pooling of tcls to reduce the spatial dimension

$$s = \text{GAP}(x) \in \text{RC}$$

- b. Channel weight w is calculated

$$\omega = \sigma(\text{Conv1 Dk (tcls)})$$

- c. Reweighted token $\bar{\text{tcls}}$ is acquired as follows:

$$\bar{\text{tcls}} = \text{tcls} \cdot \omega$$

Results

Table 6: ViT vs. SCViT accuracy (%) on remote sensing scene datasets from Lv et al. (2022).

Dataset	Train Split	ViT		SCViT	
		(%)	(%)	(%)	(%)
UCM	50%	98.140	—	98.900	—
	80%	99.010	—	99.570	—
AID	20%	94.900	—	96.560	—
	50%	96.490	—	96.980	—
NWPU	10%	91.590	—	92.720	—
	20%	93.900	—	94.660	—

Transformers in Remote Sensing: A Survey[8]

- Remote sensing data are characterized by multiple resolutions: spatial, spectral, radiometric, and temporal

Table 7: Major remote sensing modalities and key characteristics (summary from Aleissae et al. (2023)).

Modality	Description / Why it matters
Very High Resolution (VHR) imagery	High spatial resolution imagery, beneficial for land-use/land-cover understanding, change detection, object detection, and instance segmentation.
Hyperspectral Imagery (HSI)	Each pixel records a <i>continuous spectrum</i> with fine wavelength resolution (often beyond visible, spanning UV to IR). Rich spectral signatures can reflect physical structure and chemical composition.
Synthetic Aperture Radar (SAR)	Active sensing via emission/reception of electromagnetic signals. Useful for geo-localization and feature estimation (e.g., roughness, moisture, density), and for disaster management (oil slick detection, ice tracking), forestry, and hydrology.

Transformers in VHR imagery

12.2.1 Scene classification

A representative benchmark is AID (10,000 images, 30 categories), often evaluated under a train/test split such as 20/80.

Table 8: Examples of Transformer-based VHR scene classification methods (AID) as summarized in Aleissae et al. (2023).

Method	Backbone / Model	Notes	Accuracy (%)
V16_21K	ViT	ViT pretraining variant	94.970
CTNet	ResNet34 + ViT	Hybrid CNN+Transformer	96.350
TRS	TRS	Transformer-based	95.540
TSTNet	Swin-T	Hierarchical Transformer	97.200
RSP	RSP-Swin-T-E3000	Swin-based variant	96.830

Transformer in Hyperspectral Imaging(HSI)

- Frequently used in classification, often framed as assigning a category label to each pixel(or to the region around the pixel) based on spectral signatures.
- Existing work includes pure Transformer architecture, hybrid CNN-Transformers, and multimodal fusion Transformers that integrate spectral-spatial information.
- Below is the detailed explanation of HSI-BERT transformer model used for hyperspectral Imaging

HSI-BERT Model architecture

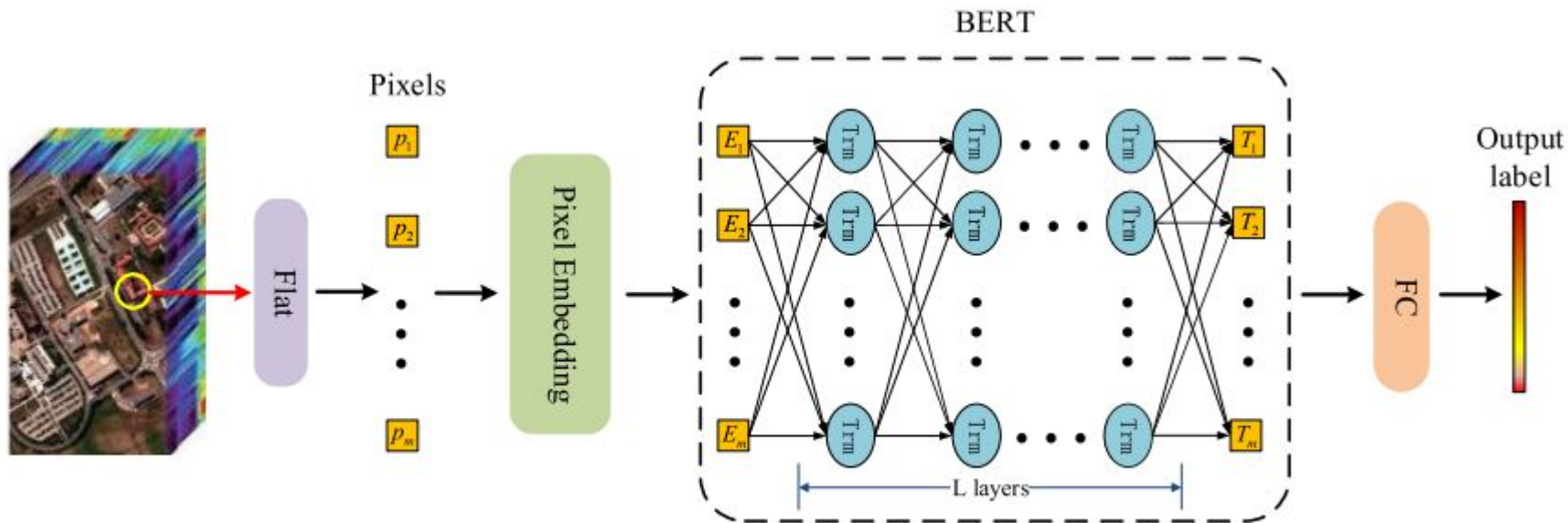


Fig. 1. Proposed architecture. $E_1, E_2, \dots, E_m$ are the embedded pixels after the pixel embedding (PE) process. Trm is the abbreviation of transformer encoder, while $T_1, T_2, \dots, T_m$ are the encoded pixel sequence processed through multiple Trm layers. FC is a fully connected layer.

HSI-BERT

Key Properties:

- Global receptive field via self-attention
- Flexible input region selection
- Joint training strategy to improve generalization.

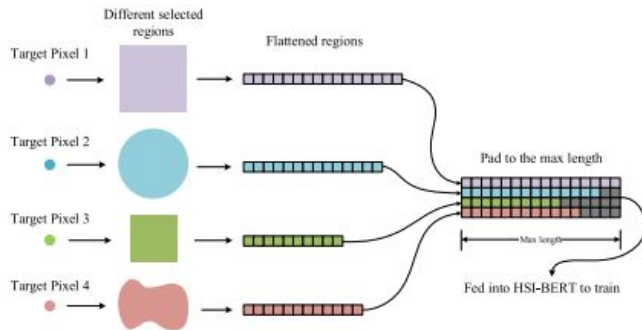


Fig. 6. Data of joint training. The “gray” pixels are dummy pixels used to pad pixel sequences to a max length. Pixel sequences from different regions are concatenated and directly fed into HSI-BERT to train.

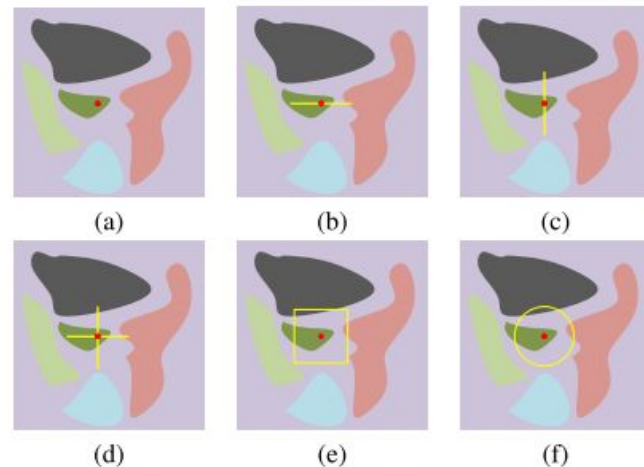


Fig. 10. Several different regions used in the experiment. The red point represents the target pixel, and the region enclosed by the yellow line together with its boundary is the selected region. (a) Dot region, contains only the target pixel. (b) Horizontal line region, with target pixel in the center. (c) Vertical line region, with target pixel in the center. (d) Cross region, with target pixel in the center. (e) Square region, with target pixel in the center. (f) Round region, with target pixel in the center.

HSI-BERT

Table 11: HSI datasets used in HSI-BERT He et al. (2020) (as described in the paper text).

Dataset	Details
Indian Pines	145 $\times$ 145 (AVIRIS sensor), 220 spectral channels (0.4–2.5 μm), spatial resolution 20 m.
University of Pavia	610 $\times$ 340, 103 spectral bands (0.43–0.86 μm), spatial resolution 1.3 m, 9 classes, 42,776 labeled pixels.
Salinas	512 $\times$ 512, 224 spectral bands, spatial resolution reported as 3.7 m, 16 classes.

Training setup:

- Number of heads = 2
- Lr = 3e-4
- Epochs = 200
- Batch size = 128
- Dropout: 0.2

HSI-BERT

12.4.3 Results

Table 12: Accuracy comparison for HSI-BERT and baselines (values taken from your notes for He et al. (2020)).

Model	Type / Backbone	Indian Pines (%)	Pavia (%)
HSI-BERT	Transformer (BERT-style)	99.560	99.750
DSS-TRM	Transformer-based	99.430	98.500
CTN	Hybrid / CNN-Transformer	99.110	97.480

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